MODULE DS.01.04 · INSTRUCTOR COMPANION

Linux Fundamentals

Speaker notes for all 23 slides — what to say, ask, demo, and watch for. Pacing plan, room checklist, and misconception watchlist included.

COURSE Data Systems Administrator

MODULE DS.01.04 — Linux Fundamentals

FORMAT SAY · ASK · DEMO · WATCH FOR, slide by slide

REHEARSE THE DEMOS · THE REST IS IN THESE PAGES

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BEFORE CLASS

How to Use These Notes

One page of setup, then a page-or-less per slide: what to say, what to ask, what to show, and what to watch for.

These notes follow the 23-slide deck *DS.01.04 — Linux Fundamentals* slide by slide. Every slide gets up to four color-coded blocks. The wording in **SAY** blocks is a suggestion, not a script — make it sound like you. Timings assume a 90-minute session: about 65 minutes of instruction plus a 25–35 minute hands-on lab.

Table 1. The four labels

Label	What it gives you	How to use it
SAY (blue)	Suggested spoken wording for the key idea	Adapt to your voice — never read it robotically
ASK (amber)	Questions to pose, with expected answers in parentheses	Pose, then wait at least 5 seconds. Silence means they're thinking
DEMO (green)	Live terminal steps, numbered in order	Rehearse once before class; narrate each step *before* you type it
WATCH FOR (red)	Misconceptions, traps, and logistics	Glance at these before each unit starts

LOGISTICS

Session at a Glance

65 minutes of instruction, then the lab. Slide numbers match the deck exactly.

Table 2. Pacing plan (90-minute session)

Block	Slides	Minutes	Core activity
Welcome & course map	1–2	4	Set expectations, terminal show-of-hands
Unit 1 — Linux Fundamentals	3–6	10	First live terminal demo: `ls`, `man`, Tab
Unit 2 — Understanding File Types	7–10	12	The rename illusion: `file notes.jpg`
Unit 3 — Create, Modify & Move Files	11–15	17	Follow-along demos: `touch` → `rm`
Unit 4 — Drives, Media & Paths	16–19	12	USB mount/eject set piece
Unit 5 — bash · CMD · PowerShell	20–22	7	Pipeline built one stage at a time
Wrap-up & homework briefing	23	3	Five practice tasks
Hands-on lab	—	25–35	Students work the five tasks; circulate

If the session runs long, compress Slides 6, 18 and 21 (talking points only) — never cut the Unit 2 rename demo or the Unit 4 mount demo; they carry the two biggest mental models.

LOGISTICS

Room & Props Checklist

Tick these off before students arrive.

- ☐ **Instructor terminal on the projector.** Linux machine, VM, WSL, or live USB — with **font size bumped up** so the back row can read commands.
- ☐ **Student terminals ready.** Lab machines booted into the prepared VM image or live USB; verify one machine per student plus two spares.
- ☐ **USB sticks.** One per student for the mount/eject demo and lab Task 4 — plus a plan to collect them at the end.
- ☐ **Pre-seeded log file.** If `/var/log/syslog` is thin on the lab image, prepare a scratch log with plenty of *error* lines for the pipes demo (Slide 22).
- ☐ **Whiteboard or flip chart.** For drawing the single filesystem tree (Slide 17) — the unit's key mental model.
- ☐ **Offline fallback checked.** `man` pages and `--help` work without internet; confirm demo files exist locally.
- ☐ **Mount demo rehearsed.** Run the Slide 19 sequence once on the demo machine — device names vary (`sdb1` vs `sda2`); never guess in front of the class.

> WATCH FOR
> The two live-demos-that-must-work are the rename/`file` illusion (Slide 8) and the USB mount/eject (Slide 19). If you rehearse nothing else, rehearse those two.

BEFORE CLASS

Misconception Watchlist

The eight ideas most likely to be wrong in your students' heads — and the slide that fixes each one.

Table 3. Misconceptions and their fixes

Students think…	Actually…	Fixed at
"Renaming a .txt to .jpg converts it"	Content never changes; Linux reads the magic bytes	Slide 8 demo
"A .tar file is compressed"	Plain .tar only bundles; the .gz compresses	Slide 9
"Scripts run themselves like .exe"	`chmod +x` and the `./` prefix do the work	Slide 9
"`>` adds text to a file"	`>` overwrites; `>>` appends	Slide 12
"Deleted files go to trash"	CLI deletion is final — no bin, no undo	Slide 14
"chmod 777 fixes permission errors"	Least privilege that works — 777 unlocks every door	Slide 15
"/root is the root of the filesystem"	`/` is the root; `/root` is the admin's home	Slide 18
"Copy dialog closed = safe to unplug"	Writes are cached; `umount` first, always	Slide 19

SLIDES 3–6 · ~10 MINUTES

Unit 1 · Linux Fundamentals

Goal: students leave knowing what Linux is, how to read a command, and where to find help.

Slide 01 · Linux Fundamentals (title)

Slide 1 of 23 · ~2 min · Welcome

SAY

"Welcome to your first step into system administration. Everything we do this course runs through one skill: telling a computer exactly what to do — one command at a time. No experience needed. Bring curiosity."

ASK

"Show of hands — who has ever typed a command into a terminal?" *(Expect few hands. Reassure the rest: that's normal, and it's why we're here.)*

WATCH FOR

Experience levels will be mixed. Set the norm on minute one: no question is too basic. Quietly note the experienced students — they make excellent lab helpers later.

Slide 02 · The Map: Five Units

Slide 2 of 23 · ~2 min · Roadmap

SAY

"Five stops. What Linux is. What file types really mean. Creating, moving and deleting files safely. How drives and paths work — no C: drive here. And how the same ideas translate across bash, CMD and PowerShell. By the end you'll complete five real tasks on a real terminal."

ASK

"What do you think a 'file type' actually is?" *(Take two guesses, don't answer — Unit 2 will surprise them.)*

WATCH FOR

Keep this slide a map, not a lesson. Don't start explaining units yet.

Slide 03 · Unit 1 Title

Slide 3 of 23 · ~30 sec · Transition

> **SAY**
> "Unit one: meet the operating system that quietly runs the internet."

Slide 04 · The Penguin Under the Hood

Slide 4 of 23 · ~4 min · Key concept + stats

> **SAY**
> "Linux is a free, open-source operating system — the basic software layer that makes a computer usable. Free twice: it costs nothing, and you're free to read, improve and share its code. Instead of clicking icons, administrators talk to it by typing short commands into a terminal — and the program that runs those commands is the shell, usually bash."

> **SAY**
> Land the stats: **100%** of the world's 500 fastest supercomputers run Linux. **3 billion+** Android devices run on a Linux kernel. Cost to download, use and share: **$0**.

> **ASK**
> "Where have you already used Linux today without knowing it?" *(Expected: Android phones, the Wi-Fi router, smart TVs, the school's website server.)*

> **WATCH FOR**
> If a student says "Linux is Ubuntu": Ubuntu is one distribution — one flavor — of Linux. Acknowledge, park it for a later module. Don't lecture on open-source licensing; one sentence on freedom is enough.

Slide 05 · Every Command Is a Sentence

Slide 5 of 23 · ~3.5 min · First live demo

> **SAY**
> "A command looks cryptic until you see the grammar: verb — what to do; option — how to do it; argument — where or to what. Read it like a sentence: list, in long format, the folder /home."

> **DEMO**

First live terminal moment — go slow: **1)** `ls -l /home` and read the output as a sentence. **2)** `man ls`, scroll once, press `q` to quit. **3)** `ls --help` for the short version. **4)** Type `ls /va`, press **Tab**, watch it complete; press Tab twice to show choices.

WATCH FOR
Some systems alias `dir` to `ls` — students who know `dir` must still learn `ls`. Man pages intimidate beginners: model them as *the* reliable source, above random forums. Check the terminal font is readable from the back row before class.

Slide 06 · Why Not Just Click?

Slide 6 of 23 · ~2 min · Motivation

SAY
"Picture a server 500 miles away: no screen, no mouse, no desktop. You type ssh user@server — and you're in. Four reasons this skill is non-negotiable: servers are headless; scripts repeat perfectly; ssh gives remote control; and it's the job — Ansible, Docker, cloud and security roles are command-line first."

ASK
"Why would a company *not* install a graphical desktop on every server?" *(Expected: wasted resources, more to break, bigger attack surface — accept any reasonable answer.)*

WATCH FOR
Counter the "CLI is outdated" attitude head-on: clicking is fine for one machine once; commands scale to a hundred machines, forever.

SLIDES 7–10 · ~12 MINUTES

Unit 2 · Understanding File Types

Goal: kill the idea that the extension defines the file — the demo, not the lecture, does the work.

Slide 07 · Unit 2 Title

Slide 7 of 23 · ~30 sec · Transition

SAY
".exe .zip .iso .sh .yaml .log — what's actually inside those letters after the dot?"

Slide 08 · The Extension Is Just a Name Tag

Slide 8 of 23 · ~5 min · Core demo

SAY
"On Windows, the extension decides what a file is. Linux doesn't trust the name — it opens the file and reads the first bytes, the magic bytes, and asks the file itself."

DEMO
The rename illusion — have students predict each result *before* you press Enter: **1)** `echo "hello" > notes.txt` **2)** `file notes.txt` → ASCII text. **3)** `mv notes.txt notes.jpg` **4)** `file notes.jpg` → *still ASCII text*. Let the room sit with that for a beat.

ASK
Slide's THINK prompt: "If Linux doesn't need extensions, why do we still use them?"
(Expected: for humans, and some applications do check them.)

WATCH FOR
This is the module's biggest misconception — renaming converts. Kill it with the demo, not with words. If asked what a 'real' photo looks like as text: open one in a text editor and show the gibberish.

Slide 09 · Run It · Pack It · Mount It

Slide 9 of 23 · ~3 min · File type trio

SAY
"Three verbs. RUN: .exe and .sh — a .sh script is just plain text full of commands. PACK: .zip

and .tar.gz — files packed like a suitcase. MOUNT: .iso — a whole disc captured in one file, ready to mount or write to USB."

DEMO

1) Write a two-line run.sh (echo something). **2)** Try ./run.sh first — permission denied. **3)** chmod +x run.sh, then ./run.sh — now it runs. The pause on 'permission denied' is the teachable moment.

WATCH FOR

Two traps from the deck: plain .tar bundles but does **not** compress — the .gz does that. And scripts don't run like .exe: the execute permission *and* the ./ prefix are both doing real work.

Slide 10 · Configs, Logs & Hidden Dots

Slide 10 of 23 · ~3.5 min · Where things live

SAY

".yaml — settings humans can read, where indentation is the grammar: spaces, never tabs. .conf and .log — settings and the system's diary, all plain text: settings in /etc, logs in /var/log. .deb and .rpm — installer packages for the package manager. And a leading dot means hidden — usually personal settings like .bashrc."

DEMO

1) ls in your home folder. **2)** ls -a — watch the dotfiles appear, point at .bashrc. **3)** tail /var/log/syslog — show the diary being written in real time.

ASK

"Why would a system hide its own settings files by default?" *(Expected: protect beginners from accidents; keep everyday listings clean.)*

WATCH FOR

Plant the YAML seed now: a single tab where spaces belong is the number-one config bug they'll hit in later modules.

SLIDES 11–15 · ~17 MINUTES

Unit 3 · Create, Modify & Move Files

Goal: students type every command in this unit — follow-along pace, not lecture pace.

Slide 11 · Unit 3 Title

Slide 11 of 23 · ~30 sec · Transition

> **SAY**
> "Touch it, name it, move it — and carefully, delete it."

Slide 12 · Make Something from Nothing

Slide 12 of 23 · ~5 min · Follow-along demo

> **DEMO**
> Students type along: **1)** `touch newfile.txt` **2)** `mkdir -p labs/week1` — explain `-p` builds the parents. **3)** `nano notes.txt`, type a line, save and exit. **4)** `echo "hi" > a.txt`, then `cat a.txt`. **5)** `echo "bye" >> a.txt`, `cat a.txt` — both lines there. **6)** Now `echo "oops" > a.txt`, `cat a.txt` — the old content is *gone*.

> **SAY**
> "The two arrows. One arrow overwrites — old content replaced. Two arrows append — new text joins the end. Same symbol family, opposite outcomes."

> **WATCH FOR**
> Pause hard on step 6 — this is the moment the `>` lesson lands. Before any redirect, have students narrate out loud which arrow they're about to use and why.

Slide 13 · mv: One Tool, Two Jobs

Slide 13 of 23 · ~3.5 min

> **SAY**
> "There is no rename command in Linux — mv does both jobs. Rename: same place, new name. Relocate: new home, same name."

> **DEMO**
> **1)** `mv old.txt new.txt` — instant. **2)** `mv notes.txt /tmp/` — relocated. **3)** Create a target that already exists and run `mv -i` — show the system asking first.

ASK
Slide's THINK prompt: "Why is renaming instant, but moving to a USB stick slow?" *(Answer: within one drive, only the label in the index changes — data never moves. Across drives it's secretly copy-plus-delete: every byte travels.)*

WATCH FOR
`mv` overwrites silently — if the target name exists, it's gone. Model `mv -i` as the seatbelt.

Slide 14 · Handle with Care: rm Is Forever

Slide 14 of 23 · ~3.5 min · Safety culture

SAY
"The command line has no recycle bin. No undo. No trash folder. GUI deletions go to trash — CLI deletions just go. And rm -rf is the chainsaw: force plus recursive, everything gone without asking. Never aim it at slash, never mix it carelessly with sudo. Professionals have wiped production systems this way — real companies, real outages."

DEMO
1) `cp -r project/ backup/` — copy before you risk. **2)** `rm -ri olddir/` — answer y and n to show the interactive rhythm. **3)** `rmdir` on an empty folder succeeds; on a non-empty one it refuses — safe by design.

WATCH FOR
Expect nervous laughter at `rm -rf` — channel it into the habit: back up first, delete second. If asked about recovering an rm'd file, the honest answer is 'backups.'

Slide 15 · Read a File's ID Card

Slide 15 of 23 · ~4 min

SAY
"ls -l is the file's ID card — learn to read every column: permissions, owner and group, size, last-modified date, and name. And the name is just a label — remember Unit 2."

DEMO
1) `ls -l` and read one line aloud, column by column. **2)** `stat run.sh` — point out the three timestamps and the **inode**: the file's true identity, unchanged by renames. **3)** `chmod u+x`

> `run.sh`, then `ls -l` again — watch the `x` appear.

> **ASK**
> "chmod 777 gives everyone everything. In the door analogy, what did you just do?" *(Left every door in the building unlocked.)*

> **WATCH FOR**
> `chmod 777` is the lazy fix that creates real security holes. Teach the rule: minimum permission that works.

SLIDES 16–19 · ~12 MINUTES

Unit 4 · Drives, Removable Media & Paths

Goal: replace the drive-letter mental model with the single tree — then make mounting a ritual.

Slide 16 · Unit 4 Title

Slide 16 of 23 · ~30 sec · Transition

SAY
"One tree. No drive letters. Everything hangs under slash."

Slide 17 · No C: Drive. One Tree.

Slide 17 of 23 · ~3.5 min · Mental model shift

SAY
"Windows gives every drive its own island with a letter. Linux has one tree: slash is the single root, and drives attach to the tree as folders. Your files keep the same address no matter which physical disk they live on."

DEMO
1) `lsblk` — read the device tree aloud: disk, partitions. **2)** `df -h` — free space in gigabytes, not bytes; explain 'human-readable.'

ASK
"If there's no E: drive, where does a USB stick show up?" *(Expected: as a folder — like /media/usb or /mnt.)*

WATCH FOR
Students will hunt for drive letters all session. Draw the tree on the whiteboard — root at top, branches for /home, /etc, /var — and point back to it whenever someone asks 'where is the D: drive?'

Slide 18 · The Map of /

Slide 18 of 23 · ~2.5 min

SAY
"/home — your files. /etc — settings for the whole system. /var — logs, the system's diary;

troubleshooting starts in /var/log. /usr — programs and shared tools. /tmp — scratch space, wiped on reboot, never store anything you care about. /bin and /sbin — the basic commands themselves."

ASK
"Something on the system breaks — which folder do you open first?" *(/var/log.)*

WATCH FOR
The trap, say it twice: /root is the administrator's home folder — it is **not** the same as /, the root of the whole tree. Same word, two different things.

Slide 19 · Plug In · Mount · Eject (+ Finding Your Way)

Slide 19 of 23 · ~5 min · Set piece demo

DEMO
The four-step ritual — narrate each step *before* running it: **1)** lsblk — spot the new device, sdb1. **2)** sudo mount /dev/sdb1 /mnt — attach the stick to a folder. **3)** cp report.pdf /mnt/ — use it like any folder. **4)** sudo umount /mnt — always eject before pulling.

SAY
"Why eject? The system caches writes — it says 'done' while the bytes are still in memory, to be fast. Pull the stick early and the last files may never arrive. umount forces every pending write to finish." Then paths: **absolute** starts from / and works anywhere; **relative** starts from where you stand. ~ is home, .. is up one level, pwd asks 'where am I?', and **Tab** finishes your sentences.

ASK
"You're standing in labs/week1. What does .. mean?" *(labs.)*

WATCH FOR
It's umount — one 'n'. Device names vary (sdb1 vs sda2): check lsblk, never guess. In the lab, students will try to skip umount — make it a graded habit in Task 4.

SLIDES 20–22 · ~7 MINUTES

Unit 5 · Commands: bash · CMD · PowerShell

Goal: show the thinking transfers — only the vocabulary changes. End on the pipe, the superpower.

Slide 20 · Unit 5 Title

Slide 20 of 23 · ~30 sec · Transition

SAY
"Three dialects of one idea — tell the computer exactly what to do."

Slide 21 · Same Task, Three Dialects

Slide 21 of 23 · ~2.5 min

SAY
"You'll meet all three shells in your career. The thinking is identical — only the vocabulary changes. ls, dir, Get-ChildItem: same job, three accents." Walk two or three rows of the table, not all of it — the table is their reference.

ASK
"Who's used CMD or PowerShell? Call out the Windows equivalent of ls." *(dir — and Get-ChildItem in PowerShell.)*

WATCH FOR
Note for the sharp students: PowerShell answers to `ls` and `cd` as aliases, but under the hood it pipes **objects**, not text. Flag it as a later-course topic — don't demo it today.

Slide 22 · Command Superpower: Read It Like a Sentence

Slide 22 of 23 · ~4 min · Build the pipeline live

DEMO
Build one stage at a time — this pacing is the whole point: **1)** `cat /var/log/syslog` — let the wall of text scroll. **2)** Add `| grep error` — only error lines remain. **3)** Add `| wc -l` — just a number. After each stage, read the pipeline aloud as a sentence: 'show me the log… keep only the error lines… count what's left.'

SAY

> "Small tools, one job each, chained with the pipe — that's the Unix way. 42 errors found in two seconds, without opening a single file in an editor."

ASK

"How would you count lines mentioning 'warning' instead?" *(Swap the grep pattern — same pipeline.)*

WATCH FOR

Students will type the whole pipeline at once, then can't locate their mistake — insist on one stage at a time, checking output after each. If syslog is thin on the lab image, switch to your pre-seeded log file.

SLIDE 23 · ~3 MIN + LAB

Wrap Up · Practice Checklist

Goal: every student starts the five tasks with confidence — and knows help is built in.

Slide 23 · Your Turn at the Keyboard

Slide 23 of 23 · ~3 min briefing + 25–35 min lab

SAY

"Five tasks, done by your own hands, in your own terminal. Read the ID cards with ls -l. Prove the name tag lies with file. Build and fill labs/week1. Do the full USB ritual — mount, copy, eject. And count the errors in a log with one pipeline. Breaking things in this lab is how admins are made — that's what the lab is for."

DEMO

No new content — walk the five tasks on the slide, then confirm every student can open a terminal *before* the timer starts.

WATCH FOR

Distribute USB sticks for Task 4 and track their return. Circulate: the first five minutes surface every 'terminal won't open' issue. Watch for students skipping `umount` — it's the habit this lab is really teaching. Remind them: `man` and `--help` are allowed and encouraged, here and in the quiz.